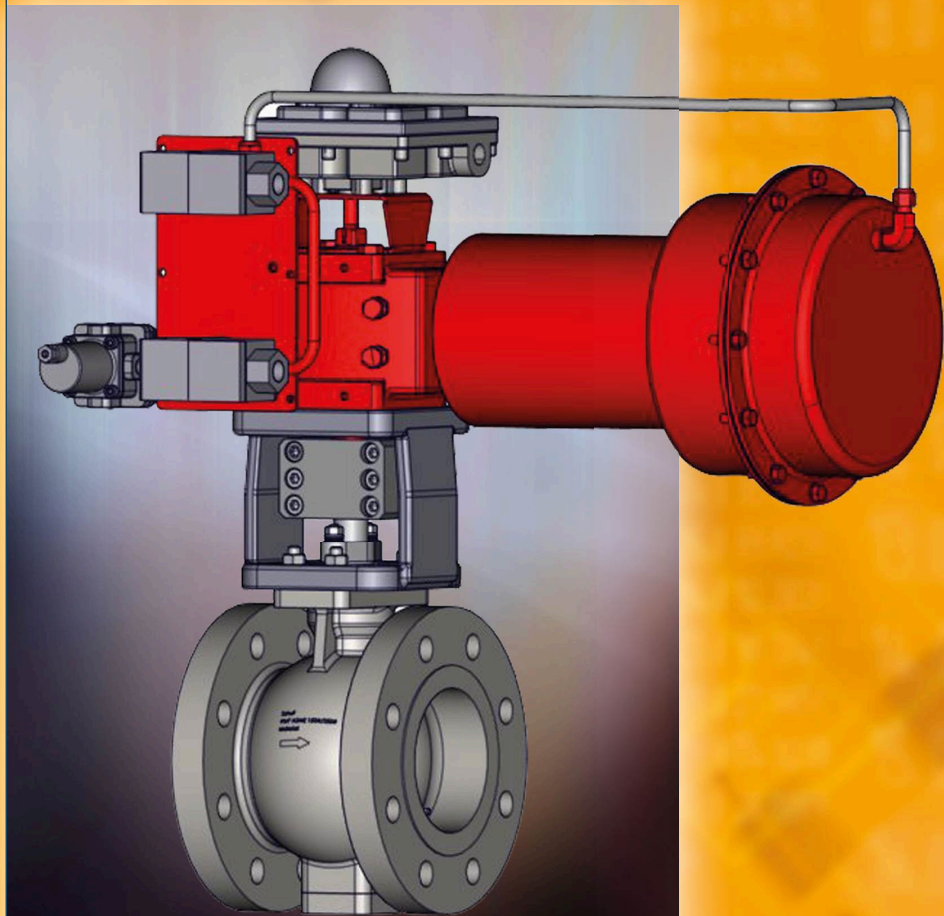


SchuF Segmented Ball Valve



SchuF 

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*SchuF is fully registered,
accredited and certified
worldwide*

SchuF Worldwide Group

SchuF is an industry-renowned valve supplier with over 100 years experience designing and manufacturing application-specific valve solutions.

SchuF invented the first bottom outlet valve over 100 years ago, and today the SchuF Group offers valve solutions in various materials and classes, with many options.

These valves are custom-manufactured to exact tolerances for use in specialised processes with often challenging operating conditions. SchuF Valves can be found in standard and severe service applications in the Chemical, Polymer, Pharma, Oil, Gas, Offshore and Refining industries.

Continuing research and development in materials and design enable SchuF to offer valve solutions for applications with high pressure, high temperature and difficult media, or a combination of all three

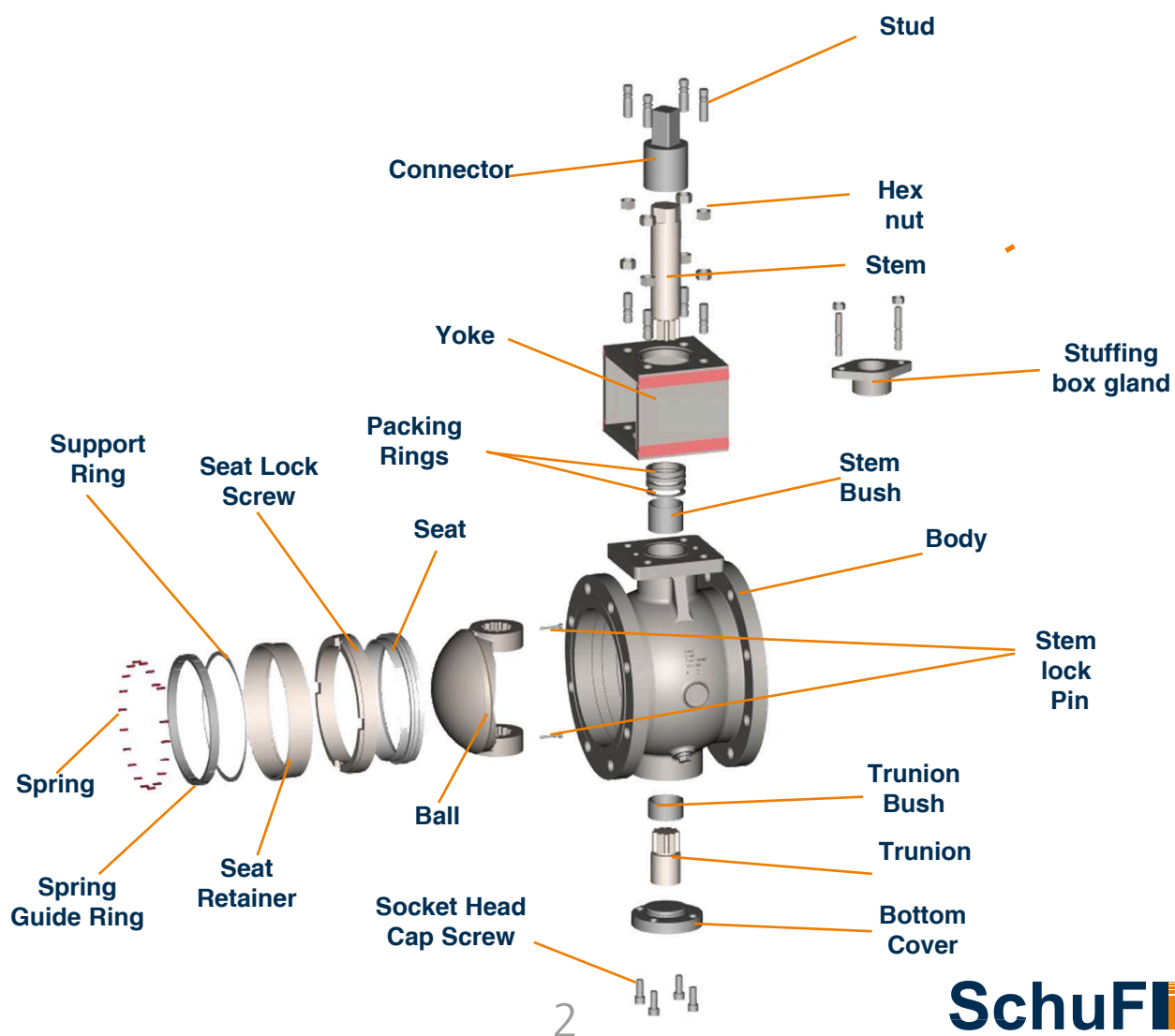
SchuF has the capability to ship our unique and highly-praised valve solutions worldwide from design and production facilities located in Germany, India, Ireland, Italy, the United Kingdom and the United States, where SchuF's skilled team of engineers and product specialists design each valve from the ground up to meet specific application requirements and provide optimal service life and performance.

Segmented Ball Valves

Segmented ball valve is a cost effective, high performance valve in a quarter turn design with metal-to-metal or metal-to-PTFE seating. It can be used as a control valve and a shut-off valve. A shearing action between the V-notch ball and the ball seal promotes smooth, nonclogging operation. The unrestricted straight-through flow design provides high capacity for gas, steam, liquids, and fibrous slurries. It is available as standard in stainless steel and carbon steel Material.

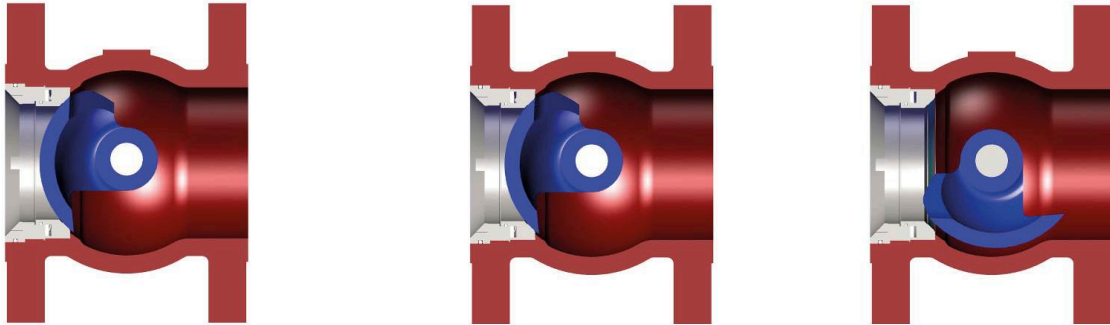
Features

- Unique stem design offers an uninterrupted flow
- Low operating torque
- Precise FlowControl.
- Customizable control.
- Tight shutoff seat design
- Self centering ball
- Noise reduction place (option available)
- Multiple trim options available for optimized performance on severe service applications
- Smooth Valve Operation
- Low emission packing
- Blowout Proof Stem
- Resists Damage from erosive flow
- Maximum Flow capacity
- Quick and easy maintenance



Double eccentric design

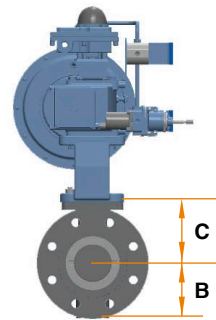
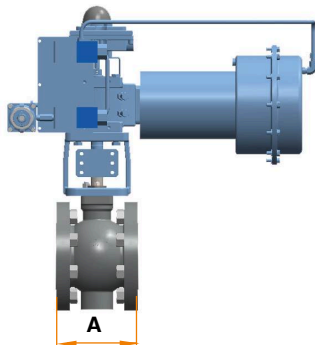
- The Ball trunion is offset from the centerline of the valve
- The face of the ball is offset from the centerline of the ball trunion



Double eccentric design Benifits

- Eliminates friction during valve open and close
- Offers more precise control
- Less wear and tear on internal parts
- Reduces the required breakway torque which inturn reduces the size of the actuator

Dimension Table: Standard Valve size, ASME 150 and 300



Valve size	Port Dia	Face to Face(A)	Center to bottom(B)	Center to top(C)	Stem Dia	Weight Kg		CV
						#150	#300	
1	19	102	77	100	12	3.7	4.7	19
2	45	124	100	130	20	8.4	9.7	101
3	73	165	117	145	26	17.3	20.3	268
4	90	194	136	165	30	26.2	33	430
6	135	229	190	220	40	43.8	57.8	880
8	170	243	222	250	50	67.5	86.5	1335
10	225	297	265	300	60	108.8	136.9	2250
12	260	338	315	350	65	167.7	205.8	3100

Notes:

1. All the dimensions to used for reference only,final drawing will be provided post order stage for approval
2. For complete Cv curves contact SchuF sales team
3. For ASME globe style valve bodies (ISA 75.08.01, EN 558 37-38)
4. If higher pressure drops are required, contact your SchuF sales office
5. Additional seat and bearing materials are available. Contact your SchuF sales office for pressure drops.
6. Values are for components shown only. Pressures/temperatures may exceed limits per ANSI B16.34 for body materials

Technical Specifications

<i>Size and Pressure Class</i>	<i>1"-12" ANSI Class 150-300 (Other sizes and rating are available on request)</i>
<i>End Connections</i>	<i>Flanged, Butt weld end and Socket weld ends</i>
<i>Design according to</i>	<i>ANSI B 16.34</i>
<i>Face to face</i>	<i>ISA-75.08.02/ISA 75.04</i>
<i>End Connections</i>	<i>ASME B16.5</i>
<i>Metal Seat:</i>	<i>Class IV per ANSI/FCI 70-2 and IEC 60534-4</i>
<i>Soft Seat:</i>	<i>Class VI per ANSI/FCI 70-2 and IEC 60534-4</i>
<i>Actuator options:</i>	<i>Pneumatic and electrical actuator</i>
<i>Characteristics</i>	<i>Equal percentage / linear</i>

Materials Of Construction

<i>Body</i>	<i>ASTM A216 Gr.WCB, ASTM A216 Gr.WCC, ASTM A351 Gr.CF8M</i>
<i>Ball:</i>	<i>ASTM A351 Gr.CF8M + Stellite 21, ASTM A479 Gr.316 + Stellite 21</i>
<i>Stem/Trunnion</i>	<i>ASTM A564 Type 630, ASTM B637 Gr.718, (ASTM A276 Type 316S as an option for upto 4")</i>
<i>Seat Ring</i>	<i>ASTM A479 Gr.316/410 + Stellite 6</i>
<i>Sealing options</i>	<i>Metal and Soft sealing</i>
<i>Soft seat</i>	<i>PTFE with 316/410 Seat Holder</i>
<i>Seat Retainer:</i>	<i>ASTM A479 Gr.316/410</i>
<i>Packing</i>	<i>Graphite</i>
<i>Bottom Cover:</i>	<i>ASTM A105</i>

Seal to Atmosphere Options

A number of options are available- stuffing-box seal for standard or high-temperature operation, plus additional live-loaded packing.

Stuffing Box

Packing rings are tightened with the stuffing-box gland to give a radial seal to atmosphere around the ram. Packing rings can be ordered in PTFE for standard operation or in graphite for high temperature applications or even in a mix of both. More specialized packing arrangements are available on request.

Live-Loaded Packing

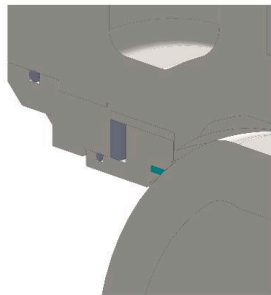
Packing rings are compressed using the stuffing-box gland, which is regularly tightened as required using standard studs/nuts. For increased service intervals, live-loaded packing, where cup-springs are employed to maintain the required minimum sealing pressure, even as the packing-rings compress over time, can provide a solution.

Seat/Sealing Options

Sealing options are available in both soft and metal designs, allowing maximum flexibility for use with all types of process

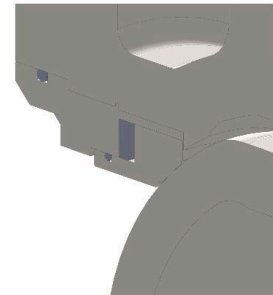
Soft- seated Valves

Soft-seated uses a thermoplastic material such as PTFE, which provides a tight seal even with low sealing force. An insert of this customer-selected sealing material is held within the seat ring.



Metal- seated Valves

Metal seated segmented ball valves use entirely metal seats with overlay / coating as per customer's choice. These also provide a long-lasting tight seal even with difficult media and a low sealing force

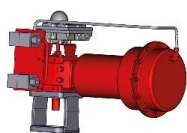
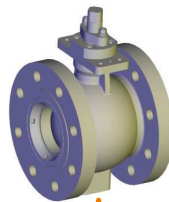


Segmented Ball Valve Applications

- Petroleum refining
- Petrochemical
- Power
- Chemicals
- Mining
- Pulp and Paper

Actuator Options

Segmented Ball Valve



Pneumatic Diaphragm Actuator



Electric Actuator



Pneumatic Single Acting Actuator

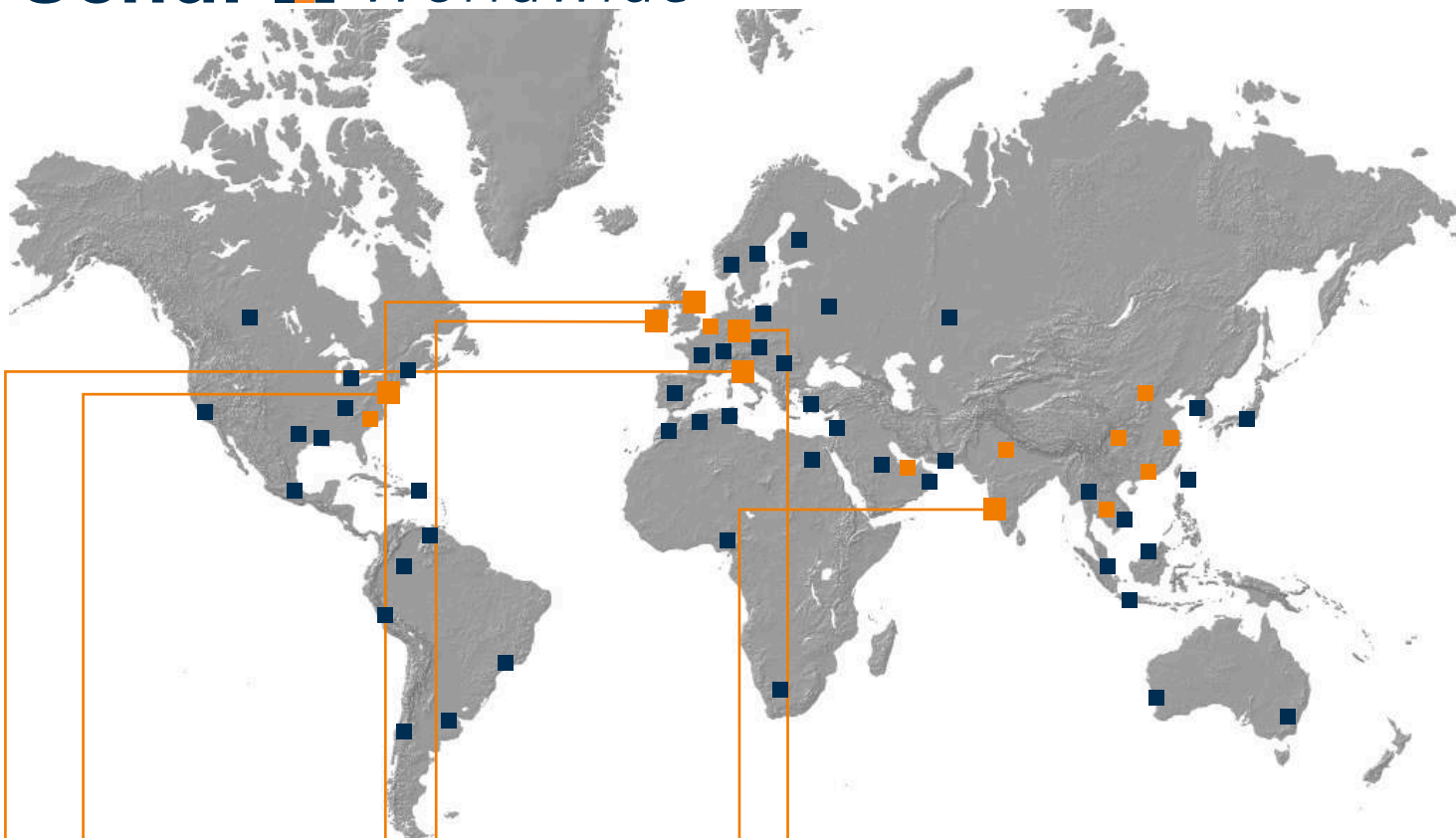


Pneumatic Double Acting Actuator

Customer Segmented Ball Valve Enquiry Sheet

SchuFI		CUSTOMER APPLICATION DATA									
		sales@schuf-india.com					www.schuf.com		Date:		
Customer							Purchase Order No				
Customer Enq. Ref.							Item Number				
Quote Reference							Quantity				
Project Title							Customer Tag				
1	PROCESS DATA	Fluid			Critical Pressure				Shut-off Pressure		
2			Units	Min	Norm	Max	Other				
3		Inlet State									
4		Flow Rate Mass									
5		Flow Rate Volume									
6		Inlet Pressure									
7		Outlet Pressure									
8		Inlet Temperature									
9		Molecular Mass									
10		Specific Heat Ratio									
11		Compressibility									
12		Specific Gravity									
13		Vapour Pressure									
14		Viscosity									
15	SPL @ 1m										
16	PIPE	Inlet Pipe Size / Sch					43	ACTUATOR	Actuator Type		
17		Outlet Pipe Size / Sch					44		Actuator Model / Stroke	/	
18		Insulation					45		Size / Spring	/	
19		Design Pressure					46		Fail Action		
20		Design Temp Min/Max.					47		System Air / Set Press.	/	
21	BODY	Valve Type					48	ACCESSORIES	Actuator Orientation		
22		Size / Class					49		Handwheel		
23		Body Type					50		Volume Tank		
24		Body Material					51		Tubing		
25		End Connections					52		Fittings		
26		End Finish					53		Stroking Time		
27	BONNET	Body F/F Dims					54	TEST	Positioner Model		
28		Bonnet Type					55		Comms Protocol		
29		Bonnet Material					56		Control Signal		
30		Bonnet Bolting					57		Switch Type		
31		Gasket Material					58		Filter Regulator		
32	TRIM	Gland Packing					59	MISC	Boosters		
33		Bellows Material					60		Solenoid		
34		Trim Size / Type					61		Lockup Valves		
35		Cv / Characteristic					62		Relief Valve		
36		Balancing					63		Other Accessories		
37	CERT	Flow Direction					64	TEST	Other Accessories		
38		Ball/Seat/Cage Matl.					65		Design and Testing: Seat Leakage, Hydrostatic, Functional Test		
39		Seating Type					66				
40	CERT	Seat Leakage					67	MISC		Paint Finish	Standard
41		PED					68		Mat. Cert Body / Bonnet	/	
42		NACE					69		Drawing Outl / GA / Hkp	No / No / No	☐
43	NOTES	ATEX					70				
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